

## Statistics Topics for Data Science (1-150)

1. What is Statistics?
2. Types of Statistics
3. Population and Sample
4. Data Types
5. Types of Variables
6. Levels of Measurement
7. Mean
8. Median
9. Mode
10. Range
11. Variance
12. Standard Deviation
13. Quartiles
14. Percentiles
15. Interquartile Range (IQR)
16. Five Number Summary
17. Skewness
18. Kurtosis
19. Coefficient of Variation
20. Histogram
21. Bar Chart
22. Pie Chart
23. Line Chart
24. Scatter Plot
25. Box Plot
26. Violin Plot
27. Density Plot
28. Heatmap
29. Pair Plot
30. Q-Q Plot
31. Basic Probability
32. Conditional Probability
33. Joint Probability
34. Marginal Probability
35. Bayes' Theorem
36. Independence
37. Random Variables

38. PMF
39. PDF
40. CDF
41. Uniform Distribution
42. Bernoulli Distribution
43. Binomial Distribution
44. Geometric Distribution
45. Negative Binomial Distribution
46. Hypergeometric Distribution
47. Poisson Distribution
48. Exponential Distribution
49. Normal Distribution
50. Log-Normal Distribution
51. Gamma Distribution
52. Beta Distribution
53. Chi-Square Distribution
54. Student's t-Distribution
55. F-Distribution
56. Weibull Distribution
57. Sampling Techniques
58. Simple Random Sampling
59. Systematic Sampling
60. Stratified Sampling
61. Cluster Sampling
62. Sampling Distribution
63. Central Limit Theorem
64. Standard Error
65. Point Estimation
66. Interval Estimation
67. Confidence Interval
68. Margin of Error
69. Maximum Likelihood Estimation
70. Null Hypothesis
71. Alternative Hypothesis
72. Significance Level
73. p-value
74. Type I Error
75. Type II Error

76. Statistical Power
77. One-tailed Test
78. Two-tailed Test
79. Z-Test
80. One Sample t-Test
81. Independent t-Test
82. Paired t-Test
83. Chi-Square Test
84. ANOVA
85. MANOVA
86. ANCOVA
87. F-Test
88. Non-parametric Tests
89. Covariance
90. Pearson Correlation
91. Spearman Correlation
92. Kendall Correlation
93. Correlation Matrix
94. Simple Linear Regression
95. Multiple Linear Regression
96. Polynomial Regression
97. Logistic Regression
98. Regression Assumptions
99. Residual Analysis
100.  $R^2$  Score
101. Adjusted  $R^2$
102. RMSE
103. MAE
104. MSE
105. Feature Scaling
106. Standardization
107. Normalization
108. Outlier Detection
109. Z-Score Method
110. IQR Method
111. Missing Value Analysis
112. Imbalanced Data Statistics
113. Trend

114. Seasonality
115. Stationarity
116. Autocorrelation
117. Partial Autocorrelation
118. Principal Component Analysis
119. Factor Analysis
120. Covariance Matrix
121. Eigenvalues & Eigenvectors
122. A/B Testing
123. Randomized Experiments
124. Control Group
125. Treatment Group
126. Blocking
127. Confounding Variables
128. Bayesian Inference
129. Prior Probability
130. Posterior Probability
131. Likelihood
132. MAP Estimation
133. Bootstrap
134. Jackknife
135. Cross Validation
136. Permutation Test
137. Law of Large Numbers
138. Markov Inequality
139. Chebyshev's Inequality
140. Jensen's Inequality
141. Entropy
142. Kullback-Leibler (KL) Divergence
143. Maximum Entropy
144. Monte Carlo Simulation
145. NumPy Statistics
146. Pandas Statistics
147. SciPy Statistics
148. Statsmodels
149. Pingouin
150. Statistical Analysis using Python